

DEBJYOTI SENGUPTA

Our Systemic World

*A Systems Theory Perspective on Humanity, Government,
Nature, and the Universe*

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“Remember, always, that everything you know,
and everything everyone knows, is only a
model.”

Donella H. Meadows

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I

THE FUNDAMENTALS

“If you can’t explain it simply, you don’t understand it well enough.”

Albert Einstein

1

Speaking in Systems: An Introduction

Systems – one of the more frequently used words across the English language, has had multiple interpretations over a period of time. The origins of the word can be traced back to the early 17th Century – appearing in late Latin and Greek, while the first usage of the word in a scientific connotation can be traced back to the 19th Century when Carnot and Clausius were laying the foundations of *thermodynamics* (the concept of understanding the behaviour of heat). The scientific connotation differs from the original Latin and Greek interpretation – “a group of principles and ideas”.

One might wonder about the need for a brief section on etymology and history above. In the modern world we currently live in, the word “**systems**” has been used to refer to administrative institutions such as the government as well as a *simplified* term for technological contraptions. There is no denying that all of humanity is surrounded and governed by **systems**. Thus, it is quintessential that we make ourselves familiar with key concepts and definitions in systems theory. Systems thinking has been used in a variety of applications and more often than not, has had different interpretations. Familiarity with a general “agreed” definition for these concepts is quite important before we indulge ourselves in their versatile applications.

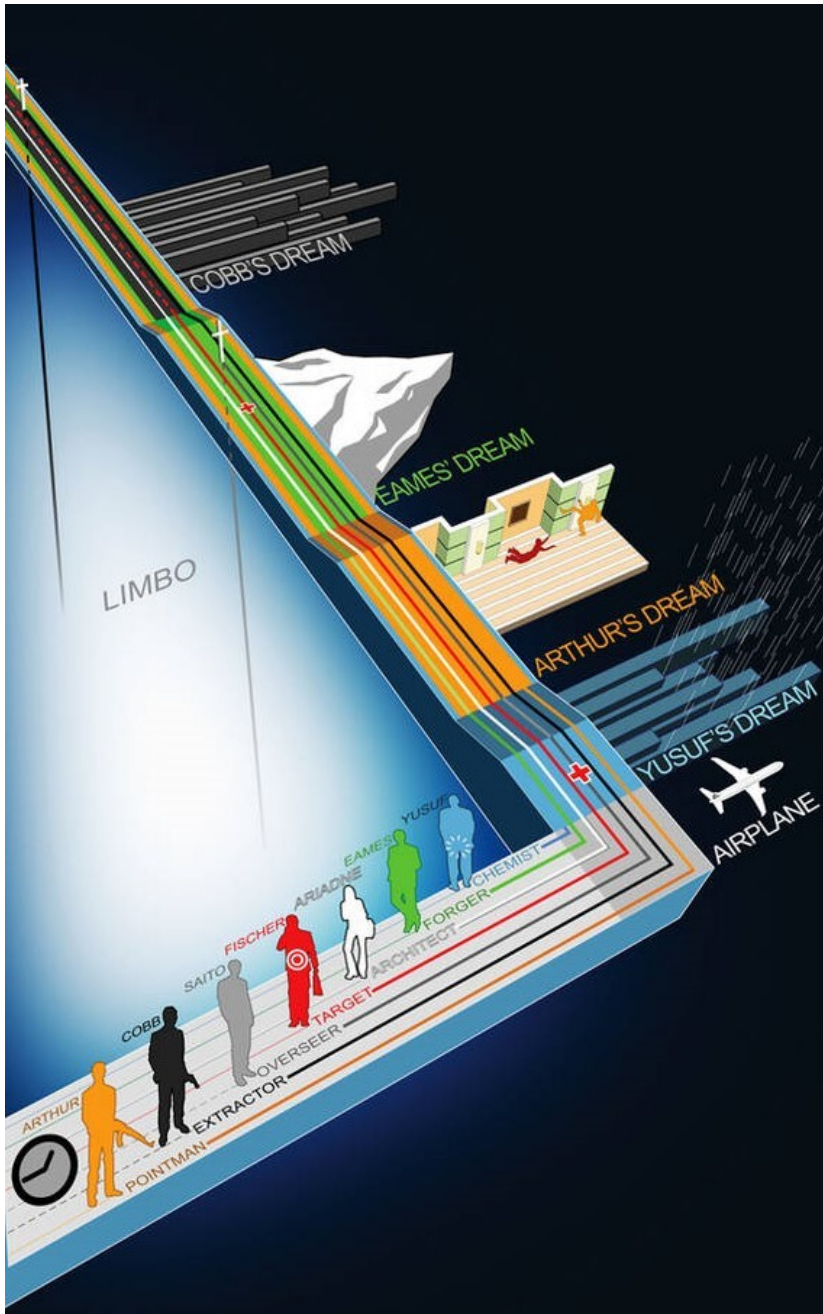
Systems

So, what is a system? The Merriam-Webster dictionary has defined it as follows:

“A regularly interacting or interdependent group of items forming a unified whole, such as:

- *a group of interacting bodies under the influence of related forces,*
- *an assemblage of substances that is in or tends to equilibrium,*
- *a group of body organs that together perform one or more vital functions,*
- *a group of related natural objects or forces,*
- *a group of devices or artificial objects or an organisation forming a network especially for distributing something or serving a common purpose,*
- *a major division of rocks usually larger than a series and including all formed during a period of era,*
- *a form of social, economic, or political organisation or practice.”*

We shall investigate the different interpretations of the word in this chapter as well as across the entire book. However, any reader would be overwhelmed by the official dictionary definition. I have tried my best to generalise this definition to the point that it can be understood by almost everyone – a system is a functioning structure based on the different constituent groups which have a dynamic relationship with each other. To further simplify this definition, let me drop a pop-culture reference from Oscar-winning director Christopher Nolan’s “*Inception*” – imagine if it were **a dream within a dream, within more dreams**. We will try to understand the movie later in the book, so this might be a good time for me to apologise for any spoilers. Still, one ought to be excited by the prospects of exploring a film plot using systems theory – an indicator of the theory’s versatility and adaptability.



A graphic showing the different “layers of dreams” and the constituent “team members” in “Inception”

If you were to find any of the definitions hard to understand, I cannot blame you since it is indeed **a complex theory** that is used as **a tool to understand, explore and simplify complex concepts**. Whether it is a conventional iPhone or a laptop you carry in your backpack, a car you drive to work, or our bodies, complex technology like the rocket, and the Universe itself – everything can be described and understood using systems theory. The idea of visualising and interpreting things that we see, hear and feel in the form of systems can often allow us mere humans to make sense of the intricate orderly chaos we experience in our lives. To describe systems theory in a nutshell – it is **the manual for your Ikea furniture in the form of nature itself**.

Now that we are familiar with what can be called a system, what about its constituents – the collection of bodies, objects, and ideas? We refer to them as subsystems – **a system within a system**. The reason behind my reference to “*Inception*” was to introduce you to this concept of subsystems since this is the beginning of the complexity. All of these subsystems have unique dynamic relationships with one another which allow the larger system to function, but remove one of these subsystems and the entire structure collapses. This is due to the inputs required for the **design of a system** – choice, demand, and constraints. A few simple examples would be as follows:

- You cannot drive a car if it does not have an engine, the radiator is damaged, or the steering and brake systems malfunction.
- A rocket would not reach its target orbit without subsystems such as the engine, the navigation computer, the fuel storage, and much more.
- A human body will be affected if a single organ is affected by illness, which is why a multiple organ failure is fatal in most cases.
- A government would cease to function in an efficient manner if any of its departments or public bodies were to cease its work, or worse, cease to exist.

All of these systems have constraints which limit them to an area where they

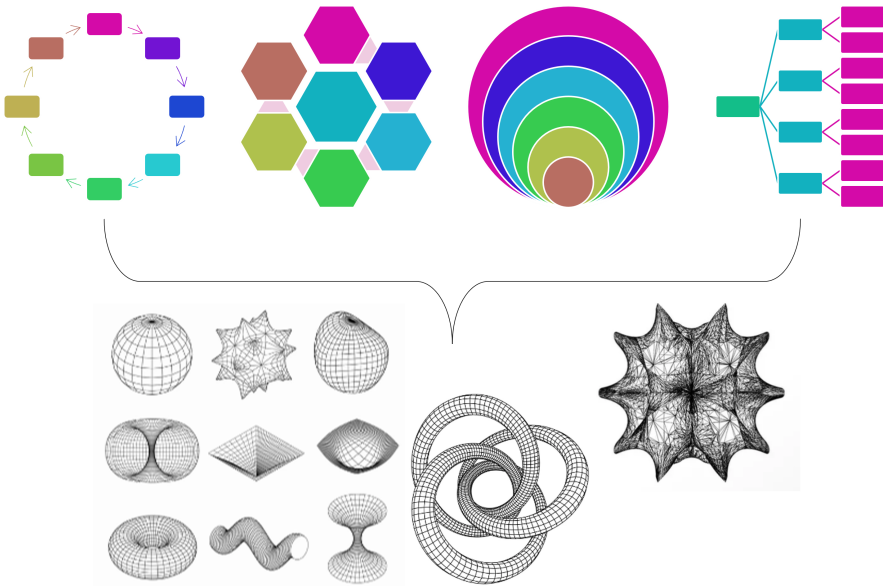
can operate legitimately, and this is valid for all systems. The demands, often referred to as **system requirements**, are required to be specific and articulate to define the purpose of a system and the tasks it is going to perform. Using the information from the demands and constraints allows a **systems designer** or a **systems engineer** to conceive a suitable concept for the design with a certain degree of choice. To summarise – the freedom in **systems design** is controlled by the constraints and demands. Most of these criteria are often set by key stakeholders who wish to use the system to achieve a certain task. For a design to be successful, the system and its features – technology, people, subsystems, relevant aspects of the physical environment – have to contribute towards achieving key objectives. One must not be misguided by the feeling that you can get your systems design spot on in the first attempt. It is always an iterative process since there is always scope for improvement and optimising the complexity of the system itself to perform more complex tasks in the constrained environment. We shall revisit these concepts when we try to understand these systems using mathematical concepts such as statistics, probability and set theory.

When we start visualising scientific and socio-economic structures using systems, we can see the importance of the fine balance and coherence required for the structure and external influencing factors. This approach has three key elements –

- a group of ideas providing a logical and philosophical rationale for the approach,
- a methodology/framework for the design, development, delivery and operations,
- a toolkit made from a collection of important principles and techniques to assist us in planning, design work and evaluation.

System Boundary, Architecture and Environment

If the previous section was meant to ease you into the concept of what qualifies as a system, this section is meant to allow one to make sense of the entropy caused by these systems inside the intricate structure created by them. These structures can take a variety of shapes and approaches – from circles to polygons, and from flowcharts to tree-like structures. Now, the human eye is limited to seeing things in a maximum of three dimensions – imagine how these structures would look in three dimensions! These structures have three key elements – **the boundary, the architecture, and the environment**.



Examples of 2D and 3D visualisations of a system architecture.

One has to define a **system boundary** before one can start visualising the architecture and the resulting solutions for systems design. This task is considered to be difficult since it is dependent on an individual's judgement – decisions about what is inside and outside that barrier between a system and

its external environment. These decisions are critical to the success of the system as the placement or delineation of the boundary is a reflection of the individual's perspective and interpretation of the problem.

Due to different biases, we are prone to making this common mistake while describing a problem – it is not solution-neutral. It is key the **problem statement** is neutral such that we can have a set of all plausible solutions. Then, we can start defining constraints which behave as optimisation parameters – allowing us to reach an optimal solution to our problem while still being aware of other solutions should the circumstances change. The lack of neutrality in a problem statement may allow us to reject a certain set of solutions at the start but it has often had terrible consequences when one does not have a backup plan prepared, or they have to go through the process of finding a solution from the beginning when time is of the essence.

Let us explore an example to further make sense of what I am describing. Following are descriptions of a problem that the energy sector often encounters:

- “Develop a strategy to ensure reliable, low-carbon energy sources meet rising global demand while reducing greenhouse gas emissions.”
- “Identify reliable, low-carbon energy solutions that can operate continuously to support a stable energy grid and help phase out fossil fuels.”
- “Create an energy solution that leverages high-output, low-carbon sources to provide continuous power and address the limitations of intermittent renewables.”
- “Propose an energy infrastructure built on dependable, high-capacity, low-carbon technologies capable of securing a sustainable future while overcoming the shortfalls of renewables.”

The International Energy Agency (IEA) has the following problem statement:

“The IEA is at the heart of global dialogue on energy, providing authoritative analysis, data, policy recommendations, and real-world solutions to help countries provide secure and sustainable energy for all.”

The question we need to ask ourselves is which of the four problem statements is the most similar to the IEA's mission? In the first instance, it would appear that all of the four statements are saying the same thing by using different words. This is the illusion that systems theory is often capable of breaking down, allowing one to *read between the lines*. Let us investigate each of the problem statements individually.

The first problem statement is the most solution-neutral of all the statements since it is open to the development of a strategy to solve the global energy crisis while moving towards the direction of zero carbon emissions. This allows us to come up with solutions ranging from concepts to methodologies, from demonstrated technologies to amendments to regulations and policy – all plausible solutions have been accounted for and the stakeholders have been made aware of the same. The second problem statement is more specific with its objectives and constraints with the use of phrases “*continuous support for energy grid*” and “*phase out fossil fuels*”, and the word “*reliable*” – narrowing down the list of solutions one has to look through an entire set.

The third and fourth problem statements further narrow the list of solutions to exploring a technology that allows us to potentially solve the global energy crisis without relying on “*intermittent renewables*” such as solar and wind energy. There are only two viable solutions to the third problem statement – nuclear energy obtained either from fission or fusion, while the final problem statement can be used to advocate for investment in nuclear fusion technologies. You can see as you progress through the problem statements, there is a sense that the objectives are becoming more specific while the constraints are piling on. The table below is a summary of the discussion.

Problem Statement	Solution Neutrality	Key Objectives	Constraints	Plausible Solutions
IEA Problem Statement	High	Provide secure, sustainable energy for all with comprehensive policy support	Environmental impact, economic viability, global accessibility	Renewable energy, nuclear fission, energy efficiency, policy reform
Problem Statement 1	High	Ensure reliable, low-carbon energy to meet rising global demand	Reduce greenhouse gas emissions, energy reliability, long-term sustainability	Mix of renewables, nuclear fission, and improved energy infrastructure
Problem Statement 2	Moderate	Identify continuous, low-carbon solutions for a stable, renewable-supported grid	Intermittency and energy storage challenges of renewables	Nuclear fission with renewables and battery storage
Problem Statement 3	Low	Develop a framework using high-output, low-carbon sources	Renewables' limitations in continuity, growing energy demands, carbon reduction targets	Expand nuclear fission, renewables with robust grid storage solutions
Problem Statement 4	Very Low	Propose infrastructure focused on high-capacity, dependable low-carbon sources	Renewables' intermittency, need for a continuous power supply, urgency in addressing climate and energy crises	Nuclear fission as primary source, supplemented by other low-carbon tech

Table 1: Analysis of the Problem Statements

From the discussion, we have demonstrated the importance of a solution-neutral problem statement and the decisions made by the systems designer to define the boundary of the system. If the boundary is required to be *redrawn*, it is easier to do so during the very early stages of development because as progress is made, the boundary is embedded into the design concept making it harder to implement any changes. Also, this exercise allows us to draw the conclusion that the purpose of the system boundary is to distinguish the *system of interest*, *enabling systems* and *systems in the operating environment* – and this is supported by one of the International Standard Organisation's standard on systems engineering¹.

When exploring the system architecture, one must realise that it is dependent on not only the boundary, but also the interconnections the system of interest has with the enabling systems and other systems in the operating environment. We cannot ignore the fact that **causality plays a key role** in keeping the structure from a catastrophic collapse. For those unfamiliar with the concept

¹ ISO 15288:2015 – 'Systems engineering – system life cycle processes'

of causality, here is a simple explanation – for any action, there is always a “why” (cause) and a “what happened” (effect). This concept can be traced back to ancient Greek philosophers such as Aristotle, but it has been made most famous by Einstein’s work on relativity. I will use the analogy of building your own personal desktop computer setup.

For a conventional desktop computer, you need the following parts – a central processing unit (also known as the processor or CPU), a motherboard, a graphics card, a cooling system, internal storage devices, RAM cards, a power supply, and few additional but optional parts. The table below explores these enabling systems and their interconnections from the perspective of causality.

Component	Description	Causal Role	Dependencies	Effects
CPU (Central Processing Unit)	The "brain" of the computer, responsible for processing instructions and performing calculations.	Primary processing unit driving overall system operations.	Dependent on power supply, cooling system, RAM, and motherboard.	Directs overall system operations, influences the performance of all connected components.
Motherboard	The main circuit board that connects all components, enabling communication among them.	Acts as a central platform for component interconnections and data flow.	Dependent on CPU, power supply, and RAM for basic functionality.	Facilitates communication and data transfer between components, enabling full system function.
Graphics Card (GPU)	Processes visual data and renders images, graphics, and videos.	Enhances visual processing capability, crucial for tasks like gaming and design.	Dependent on CPU, power supply, motherboard, and cooling system.	Generates visual output; supports high-resolution display and accelerates graphical tasks.
Cooling System	Keeps the CPU, GPU, and other components within safe temperature ranges.	Maintains system stability by preventing overheating.	Dependent on power supply and components producing heat (CPU, GPU).	Prevents component damage, enhances system longevity, and stabilizes performance.
Internal Storage (SSD/HDD)	Stores the operating system, software, and files permanently.	Provides long-term data storage, enabling data retrieval and application loading.	Dependent on power supply and motherboard for connectivity and data transfer.	Supplies data to CPU and RAM, affecting load times and data access speed.
RAM (Random Access Memory)	Temporary storage for active processes and data, facilitating fast access for the CPU.	Holds active data to speed up processing; enhances system responsiveness.	Dependent on CPU, motherboard, and power supply for operation and data flow.	Reduces processing lag, speeds up data access for CPU, and improves multitasking capabilities.
Power Supply Unit (PSU)	Provides electricity to all computer components.	Supplies power necessary for component function and overall system stability.	Dependent on stable electricity input.	Powers and stabilizes all other components, ensuring operational continuity.
Optional Parts (e.g., Sound Card, Network Card)	Provide additional functions like enhanced audio and internet connectivity.	Expand the system's functionality based on user needs.	Dependent on motherboard for connectivity, CPU for processing, and power supply.	Enable specialized tasks, improve user experience, and enhance functionality.

Table 2: Causality analysis of parts in a desktop computer

We can interpret the information in the table using the following example – if the visual graphics on the computer is the effect, the graphics card is the cause. You can clearly see that the computer will not work properly in the way we want if one of these parts are missing due to their relationships with one another. Surprisingly, we have not talked about the operating system

(OS) of the computer since it acts as a boundary for the system of interest (computer hardware) while it acts as an enabling system for the components. The interconnecting relationships of these parts are allowing the system of interest, the computer, to function in the way we have specified using a solution-neutral problem statement. These are the elements a designer, an engineer, or a thinker have to keep in mind while proposing a solution for any problem using systems theory since they are key to the system architecture. Terms like *architecture*, *framework* and *solution* have been loosely used to describe the same thing whereas we can see from the discussion that the three are distinctly different from each other. Let us explore the system architecture of a public organisation like government administration consisting of several groups of individuals.

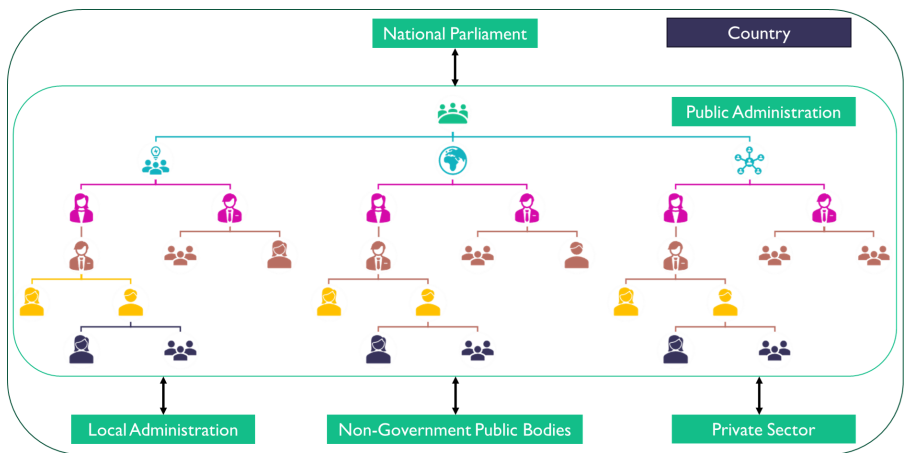


Illustration of a public organisation (such as the government) using systems theory

In a democracy, the country and the common mass – the electorate – elect individuals who will represent the voice of the nation in the Parliament. These *members of Parliament* are answerable to the nation while taking advice on implementing their plans and policies through various departments of the Civil Service, which act as enabling systems. These various departments are subject

to reorganisation since different administrations, led by either a President, or a Prime Minister (or the equivalent, or both) want to deliver different mission statements. Under such circumstances, different individuals are recruited to perform different functions within the structure. Most of you, the readers, should be able to relate to the previous sentence regardless if you are affiliated with the private sector or the public sector. If you are able to relate, you will probably think of the people you often interact at work. Each conversation you are having at work is unique, but having the potential for some overlap which allows you to work in teams. The entire public administration setup as shown in the figure has relationships with other enabling systems like local administrations, public bodies not under the government's direct supervision, the private sector and charities – working collectively as a system of interest to deliver key objectives for progress of the nation (the operating environment). Well, there you have it – you can explain bureaucracy using systems theory!

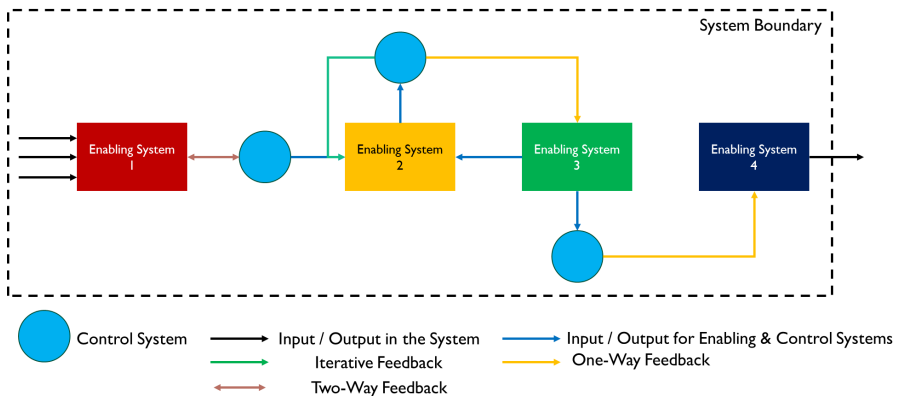
Control - An Illusive Objective

Often, we wonder to ourselves wishing we had more control in our lives. Scientists wish they had more control on variables in an experiment in their quest for perfection. Financial gurus and bankers hope they had more control on the stock markets and stabilise it. This observed continuum in the idea of control is quite fascinating if we were to view the above statements from a systems perspective. However, it also reinforces the Einsteinian belief that it is a persistent illusion. Just like we have law enforcement agencies and a judiciary structure, systems have what we refer to as “*control systems*.”

The purpose of these control systems is to prevent the entire system behaving erratically at random by checking against a set of parameters defined by stakeholders or the system designer. As it is obvious by now, designing these control systems is a difficult task as the number of constraints typically outnumber any other information essential to narrow down to an optimal solution. The control systems also allow us to troubleshoot, if the system

going into an unpredictable behaviour could not be prevented. The desire of having control of these systems can be attributed to the foresight shown by the thinkers – sometimes the random behaviour may have minimal impact, but catastrophic in other instances. This could be best explained through the example of the Chernobyl nuclear disaster in 1986 – how did one of the then “safest” nuclear reactors in the Soviet Union explode? The answer lies in two parts – human mismanagement through the failure in implementing suitable control measures, and failure in control systems which made the reactor into a bomb resulting in a catastrophe which rendered an entire town unsafe for human habitation for several millennia.

To visualise these control systems, we often use “flow diagrams” for simple as well as complex systems. The only difference between the two is we can draw the diagram by ourselves for simple applications while we need a computer (sometimes, even a supercomputer) for complex applications. Still, we need to know the *language* used for understanding these diagrams – *the language of arrows*.



A flow diagram visualising a system design

Using the legend which labels the different arrows seen in the above flow

diagram, let's try understanding the system. Our first observation starts with the fact that the system within the system boundary has four enabling systems and three control systems – all having different functions. The system boundary allows us to visualise the interface the system has with the external or operating environment. The first enabling system accepts user-defined inputs and constraints to perform a particular function within the system. The enabling system has a *two-way feedback interface* with one of the control systems, but what does this mean? Interfaces are described as the architecture which supports the connection between the two systems, and a two-way feedback means the control system processes the input from the first enabling system and sends data back to it for user visualisation while triggering the chain of tasks or commands needed for the other enabling systems to deliver their functions. Going back to the illustration, different arrows have been used to show different forms of feedback and relations with the enabling and control systems. The illustration can be interpreted as follows:

1. The first enabling system accepts the user input, performs its functions and outputs to the first control system.
2. The first control system feeds back processed information back to the first enabling system for user visualisation after checking the input received satisfies certain control conditions. It also sends the processed information to the second enabling system.
3. Enabling Systems 2 and 3 and the second control system work in tandem with one another, creating an iterative system. The second control system has two sets of control conditions – one to continue the iteration process, and one to end the iteration process.
4. The third control system receives input from Enabling System 3 and sends processed information as feedback to the Enabling System 4 which delivers output to relevant stakeholders.

The illustration and the explanation above is a suitable example of the importance of understanding how systems communicate with one another. Also, this illustration is a sample representation of a computational algorithm

for the optimisation of an engineering output like the design of a car or a bionic arm. Before we end this discussion and delve ourselves into the finer details and advanced concepts, I would like to highlight one twist about the diagram – this diagram is multi-purpose. You must be wondering “why?” Even though it can be used during the operation of the final system by relevant stakeholders, the diagram is a visualisation often employed by systems designer to verify if the concept is valid before they build the system. We shall discuss how system validation is a key principle for integration later. However, for the time being, I can proudly say that most readers are equipped with a vital component to understanding systems theory in its truest sense through mathematics and its applications – the language of systems.

2

Mathematical Mischief: Set Theory in System Design

We know that systems theory allows us to understand the constructs made by interconnected systems and how these systems behave, regardless of whether these systems are natural, engineered or social. These systems have some notable similarities to the mathematical concepts of **sets**. Reminiscing the days of your school when you had your respective mathematics classes, there must be a few who revelled in this interesting concept while others who have an utter disdain - partly to do with your understanding of the theory behind sets and partly to do with your individual opinions of your “scary” mathematics teachers. In my humblest opinion, if systems theory were a painting then set theory can often be considered as the master painter. Let us revisit the key takeaways from the first chapter -

- Boundaries define the limits of what is included while components (or the enabling systems) are individual elements of the system of interest.
- The relationships between the enabling systems determine the behaviour of the system under investigation while feedback and control systems determine the dynamics of the system.
- Behaviours and properties of a system cannot be attributed to a single enabling component or system due to the complex interfaces and dynamics.

If we are to truly appreciate the chaotic beauty of systems, it is necessary that we learn about these sets which are “collections of different things” and their intricacies. I have tried my best in giving a brief overview of the **set theory** before we delve into discussing the similarities, differences and unique qualities. The one-line explanation for the transposition from a discussion about systems to a mathematics lecture on sets is quite simple - the language used in both theories are similar in nature.

We have already described a set as a group of things. The things which are part of the set are referred to as elements. The conventional way of representing a set is as follows:

$$\mathbf{S} = \{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \dots, \mathbf{E}_{i-1}, \mathbf{E}_i\} \in \mathbf{C}$$

where **S** resembles the set we are investigating, **E** resembles the **elements** inside the set, the Greek symbol $\in$ is mathematical notation for the operation “belongs to”, and **C** is the set of user-defined **constraints** (also referred to as the **domain**). In the notation, you may notice the subscript running in numerical order from 1 to *i*. In this context, the letter *i* indicates the total number of elements in the set. Thus, the notation $\mathbf{E}_1$ would mean it is the **first element of the set**. Likewise, $\mathbf{E}_i$ would mean it is the **last element of the set**.

It is obvious that the mathematical notations could be quite difficult to interpret - more so if there is a presence of Greek letters, calculus signs and complex symbols. However, we can understand the notation better if we use the English language to our advantage in this brief fun exercise -

- Words and phrases starting with the alphabet ‘**S**’ = Sets, Systems
- Words and phrases starting with the alphabet ‘**E**’ = Elements, Enabling Systems
- Words and phrases starting with the alphabet ‘**C**’ = Constraints, Conditions

In the very first instance, we can note the similarities that systems theory and

sets theory not only in the language used to understand the mathematical notations but similar concepts and themes behind them. I have explored the definitions of a set and a system – let us see if we notice any differences!

1. “A **set** is defined as a collection of distinct **elements** that are grouped together according to specified **constraints**.”
2. “A **system** can be defined as a set of interconnected **enabling systems** and components that interact under specific **conditions** to achieve a common goal or function.”

Some of the world’s most powerful AIs (artificial intelligences) and reputable similarity checking software would give the two statements a similarity score varying from 60% to 80%. This is to do with the two statements sharing a similar theme and trying to convey similar ideas for two “different” objects. However, this is not sufficient to answer the golden question – **are sets and systems two sides of the same coin?**

Basic Principles of Set Theory

We have opened ourselves up to the concepts of sets and their constituent elements. Now, it is important to know about some of the basic properties of a set –

- A set can be part of a larger set. Under such circumstances, we can consider the larger set as the **superset** of the original set. The original set is then referred to as the **subset** of the large superset. If we are to consider **each element as an individual set**, then these sets ($E_1, E_2, E_3, \dots$) are subsets of the superset **S**. The symbols $\subset$ and $\supset$ are used to indicate the relationships of subset and superset between the two sets respectively. The following examples will help us interpret the symbols:

$E \subset S$: The set **E is a subset of set **S****

$S \supset U$: The set **S** is a **superset** of set **U**

- Just like numbers, you can “add” and “subtract” sets. These operations are referred to as **the union** and **the intersection** respectively. The union of two sets is another set which consists of elements from the two original constituent sets, while the intersection is referred to as the set of elements which are common between the two original sets. The symbols $\cup$ and $\cap$ are used to denote the union and intersection of two sets respectively.

$$S_1 \cup S_2 = \{E_{S11}, E_{S21}, E_{S12}, E_{S22}, \dots, E_{S1i}, E_{S2i}\}$$

$$S_1 \cap S_2 = \{E_1, E_2, \dots, E_i\} \text{ if } E_1, E_2, \dots, E_i \text{ are elements present in sets } S_1 \text{ and } S_2$$

- The environment around the set can be expressed as the **set difference** or the **complement of the set relative to the environment**. The comprising data contained in the resulting set in theory is **infinite**, but primarily is limited to **parameters and constraints that control the external environment**. Often, the environment is represented using a **universal set** denoted using the letter **U**. The resulting complement of a set in the universal set is referred to as the **absolute complement**. The corresponding mathematical notation for the concept is

$$U \setminus S_1 = S_1^c = \text{Absolute Complement of the set } S_1$$

- All sets have a **power set** which contains all possible **combinations** of subsets of a given set. Therefore, the power set for $\mathcal{P}(S)$ if the set had 2 elements would be all three possible combinations of subsets - $\{E_1\}$, $\{E_2\}$, and $\{E_1, E_2\}$.

Through the Lens of Logic

Before we delve into exploring the similarities between sets and systems, it must be known that we are still not well-equipped to truly appreciate systems theory through sets since concepts such as Venn diagrams, fuzzy logic and

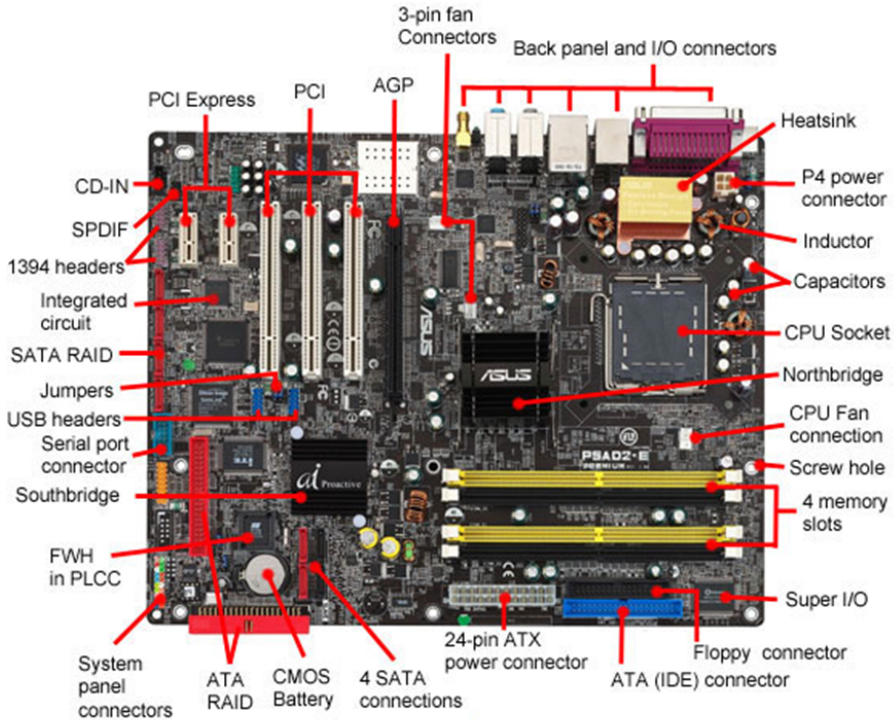
graph theory have not been explored yet. Also, if we are talking about sets – the concept of probability, uncertainty and statistics also comes into play. While we are aware of our current limitations as readers, we can still identify extensive similarities between the two objects of interest.

Focus on Relationships

Both theories emphasize relationships between components. In set theory, this is captured through unions, intersections, and mappings, while systems theory examines inter-component interactions. In this image of a computer motherboard, you can clearly see that it requires several “elements” (or components) to make the motherboard functional and it needs to interface with the CPU (the processor) and other vital components such as cables such that it behaves as a computer system. Referring to our original notation for sets,

$$\mathbf{S} = \{E_1, E_2, E_3, \dots, E_{i-1}, E_i\} \in \mathbf{C}$$

components such as CPU Socket, Integrated Circuit, CMOS Battery and others can be considered as **elements of the set S**. Meanwhile, the motherboard serves as the suitable parallel for the set **S**. This goes without saying but if the components are the elements, then the union of these elements would be the motherboard set. From the description, we can correlate that **the computer system acts as the universal set**. An interesting area to discuss would be the set **C** – what makes up the constraints for the design for a motherboard? Besides geometrical and financial constraints, the different ports like PCI, 3-pin fan connectors, power connector serve as the key design constraints since **these connection points are the interfaces (or intersections, in the case of sets)**.



A typical motherboard of a computer

Structural Representation of Hierarchies and Subdivision

Systems theory and set theory converge in their ability to represent and analyse hierarchical structures and subdivisions, offering frameworks to deconstruct complex systems into organized and understandable components. In systems theory, hierarchies illustrate the layered organization of a system, where each level represents a subset of the whole, encompassing elements with specific roles or attributes. For instance, in an ecological system, the hierarchy ranges from ecosystems to communities, populations, and individual organisms. Each level interacts within its tier while contributing

to the broader system. Similarly, set theory employs subsets to represent hierarchical relationships within a universal set. These subsets define smaller, specific groups of elements, maintaining their connection to the overall set while enabling focused analysis of their properties and roles. This structural correspondence highlights the shared capability of both theories to model complexity through systematic subdivision.

As we saw earlier, at the core of systems theory is **the definition of boundaries**, which separates a system from its environment. This concept mirrors set theory's distinction between sets and subsets, where the boundaries of a subset determine its membership and scope within the universal set. This parallel is particularly evident in nested systems, where subsystems exist within larger systems, akin to subsets nested within larger sets. For example, a corporate organization can be viewed as a hierarchical system comprising departments, teams, and individuals. Each of these components can be analysed independently as distinct subsets or collectively as part of the entire set. By capturing these layers of subdivision, both systems theory and set theory offer structured approaches to represent the intricate organization of real-world systems.

This shared focus on structural representation is instrumental in modelling nested hierarchies and their subdivisions. In a hierarchical system, each level or subset has its distinct role, but its existence and function are defined by its relationship to the broader system. In set theory, subsets similarly derive their meaning from their inclusion in the universal set, enabling the modelling of hierarchical relationships such as membership, inclusion, and exclusivity. For instance, in a government structure, a subset representing local governance is part of a larger set that includes regional and national governance. Systems theory and set theory together provide a means to formalize these relationships, offering clarity in understanding how smaller units contribute to and interact within a larger whole.

Abstraction and Universality

Both systems theory and set theory also **emphasize abstraction and generalization, making them applicable across diverse domains**. Systems theory generalizes the concept of hierarchies and subdivisions to model phenomena in fields such as engineering, biology, sociology, and economics. Set theory, being foundational to mathematics, finds applications in logic, computer science, and data analysis. This shared versatility stems from their reliance on fundamental principles: defining components, relationships, and hierarchies. For example, in project management, a work breakdown structure—a hierarchical representation of tasks and subtasks—can be modelled as a system of interrelated subsets, demonstrating the utility of these theories in practical settings.

Systems theory and set theory share a profound similarity in their emphasis on abstraction and universality, allowing them to transcend specific domains and provide a framework for analysing and understanding complex phenomena across diverse disciplines. Both theories are grounded in the idea of distilling complex entities into generalized structures that can be universally applied, making them powerful tools for representing relationships, hierarchies, and interactions in both natural and artificial systems. Through their shared focus on abstraction, these disciplines reveal common patterns that underpin seemingly unrelated systems, enabling insights that are both theoretical and practical.

Abstraction in systems theory involves reducing a system to its essential components and their interactions, stripping away extraneous details to focus on the **core dynamics and structure**. This process allows systems to be represented as models that are not tied to specific instances but can apply across various fields. For instance, the concept of feedback loops is a **universal principle in systems theory**, relevant to biological homeostasis, engineering control systems, and economic market behaviour. Similarly, set theory abstracts mathematical and logical relationships, describing the

fundamental properties of collections of objects without regard to the nature of those objects. A set could represent numbers, points in space, or even abstract concepts, demonstrating its universality as a foundational framework.

Both systems theory and set theory achieve universality by emphasizing relationships over specifics. Systems theory models relationships among components using concepts such as inputs, outputs, and flows of information or energy. These relationships are expressed through diagrams, equations, or qualitative descriptions that capture the essence of how the system functions. Set theory, meanwhile, formalizes relationships between sets, subsets, and their elements using operations like union, intersection, and complement. These operations provide a universal language for describing how different parts of a system relate to one another, enabling applications from database design to network analysis.

The universality of systems theory and set theory makes them versatile tools for interdisciplinary applications. Systems theory has been used to model ecosystems, social dynamics, industrial processes, and even philosophical ideas about interconnectedness and emergence. Its abstractions, such as the idea of systems boundaries or open versus closed systems, allow it to adapt to the unique demands of different disciplines. Likewise, set theory serves as the bedrock for much of modern mathematics, computer science, and logic. Its abstraction of elements and relationships supports the development of fields as diverse as artificial intelligence, topology, and information theory.

Another key aspect of their abstraction and universality is **scalability**. Systems theory can be applied at various levels of complexity, from small-scale subsystems to global networks, maintaining its principles regardless of scale. Set theory similarly scales seamlessly, whether analysing finite sets of discrete objects or infinite sets in theoretical mathematics. **This scalability ensures that both theories remain relevant and adaptable to problems of any size or scope.**

Diagrams, Diagrams, and More Diagrams!

Despite the key similarities between the two, system theory and set theory do have points where they diverge due to limitations on either. The two key differences are that systems theory incorporates the concepts of dynamics, feedback and emergence – ideas not dealt by the set theory. This can be attributed to the set theory’s limitation with static representation and the complex interactions observed in systems theory. Thus, the need for more advanced tools and extensions to allow us interpret systems theory using sets. These advanced tools range from visual diagrams such as Euler–Venn diagrams to complex ideas such as fuzzy logic and graph theory.

Fuzzy logic extends classical set theory by accommodating degrees of membership, offering nuanced modelling for systems with ambiguity or imprecision. Graph theory provides a robust framework for analysing relationships and structures within systems, such as networks or feedback loops. Euler–Venn diagrams, meanwhile, offer intuitive visual representations that bridge the gap between theoretical constructs and practical applications. Together, these concepts enrich the analytical toolbox, enabling a more comprehensive understanding of complex systems and their interrelationships through the lens of set theory.

Where Worlds Collide: Exploring Venn Diagrams

Venn Diagrams are by far the most basic tool to visualise the **unions, intersections and other vital relationships** formed between two or more sets. As the number of sets start growing, visualising these relationships becomes a more complex task. Mathematicians often employ the assistance of complex shapes and computational assistance to visualise these diagrams. Nevertheless, one cannot deny the intricate beauty of these diagrams when they take shape while revealing the “bigger picture”.

Time for Fuzziness: When a “Yes” isn’t always a “Yes”

To prepare ourselves for the mind-boggling uncertainty we are about to explore, I want all the readers to do a few training exercises which physicists refer to as “*thought experiments*.” Luckily, these thought experiment do not require you to have a primer course on modern physics or force you into buying a train ticket just to get a first-hand experience. Speaking of trains, this first thought experiment of mine does involve you, the reader, sitting in a train

Imagine yourself sitting in the window seat of a high speed train, admiring the beautiful sceneries on the other side of the window. However, as the passenger of a train on an unfamiliar route – you don’t know the speed of the train and you don’t know where you are. After opening up your mind to all possibilities by momentarily shutting down all your biases, focus long enough on the outside scenery and not the window. When you start looking at outside for a considerable time, try focussing on one particular object like a house, tree, or a rock. Now, ask yourself the question – if you know the location of the rock, do you know its speed or if you know the speed of the rock, can you precisely determine the location? After you have done that, focus on the window for long enough and you will notice the outside scenery being really hazy. Now, ask yourself the same questions as you did in the previous case. I am certain majority of the readers will feel “*fuzzy*” from this exercise since you do not have clear and precise answers to the questions in either of the cases.

The fuzziness results from the fact that our brains have a hard time accepting **the uncertainty associated with the quantum world**. Humans have always had a deep-rooted desire to categorize and define. Whether it is understanding the precise location of an object or its motion through space, we often find comfort in clarity. Yet, the quantum world defies this clarity, challenging the very foundations of how we think about reality. This fuzziness, however, is not a limitation—it is a doorway to understanding something profound about the nature of systems and the frameworks we use to comprehend them.

In **systems theory**, much like in quantum mechanics, ambiguity and inter-connectedness are not anomalies but integral features. This brings us to an important question: *can the fuzzy boundaries we observe in thought experiments and quantum mechanics teach us something about the complexity of systems in the real world?* The answer lies in a deeper exploration of **set theory**, which is fundamental to systems theory.

Traditional set theory operates on the principle of **binary inclusion**: an element either belongs to a set or it doesn't. This crisp distinction mirrors our classical understanding of systems, where components and their roles are neatly defined. However, in the real world, systems are rarely this straightforward. Their boundaries are often blurred, their components interdependent, and their behaviour influenced by external factors in ways that cannot be precisely delineated.

To address this, extensions of set theory, such as **fuzzy set theory**, were developed. Unlike classical sets, where membership is an absolute truth, fuzzy sets allow elements to belong to a set with varying degrees of membership. For instance, consider the system of a "forest." In classical terms, a tree either belongs to the forest or it doesn't. But in fuzzy terms, a tree on the periphery of the forest may belong to it partially, depending on its proximity, ecological role, or visual context.

This fuzzy logic aligns closely with the uncertainties of the train window thought experiment. When we fixate on the "location" of a tree, its "speed" becomes ambiguous, and when we focus on the speed, the location becomes hazy. Similarly, when analysing a system, focusing too intently on one aspect often obscures others. This interplay of focus and context highlights the **multi-dimensional complexity of systems**.

Acknowledging fuzziness doesn't mean abandoning precision—it means embracing the nuanced interplay of clarity and ambiguity. In systems theory, this leads to a **paradigm shift**: from seeking absolute answers to

understanding probabilistic trends and interdependent variables. This shift is critical in addressing **real-world challenges**, from climate change to economic instability, where clear-cut solutions are rare, and systemic interactions dominate.

Much like quantum mechanics challenges us to rethink our understanding of particles and waves, systems theory—augmented by extensions like fuzzy logic—challenges us to rethink our understanding of structure and function. By adopting a perspective that accommodates uncertainty, we not only gain deeper insights into the behaviour of complex systems but also equip ourselves to design frameworks that are **adaptive, resilient, and inclusive** of the fuzziness inherent in our systemic world.

The train journey, then, is not just a metaphor for quantum uncertainty but a lens through which we can begin to appreciate the **fluid boundaries of systems**. It invites us to accept that a “yes” isn’t always a “yes,” and a “no” isn’t always a “no.” Instead, it challenges us to think in terms of degrees, probabilities, and interconnected possibilities—a mindset that lies at the heart of both quantum mechanics and systems theory.

Networks and Narratives: Power of Graphs and Categories

In our exploration of systems theory, we have seen how interconnected systems form a tapestry of components, boundaries, relationships, and dynamics. Like a symphony composed of many distinct instruments, the beauty of a system emerges from the interactions of its parts, not the individual notes. In the language of mathematics, these interactions and relationships can be elegantly described using **graph theory** and **category theory**, which are natural extensions of set theory. Together, they provide a powerful framework for understanding the complexities of systems.

Let us begin with graph theory. If a system can be thought of as a set of enabling

systems, **graphs** allow us to describe the relationships between these systems. A **graph** is a mathematical structure comprising nodes (or vertices) and edges (or connections). Formally, a graph **G** can be written as:

$$\mathbf{G} = (\mathbf{S}, \mathbf{R})$$

where **S** is the set of subsystems or the components of the **system of interest**, and **R** is the set of the relationships or interactions between the components, often referred to as **edges**. Consider a simple example of an ecosystem: the nodes could represent species, and the edges represent interactions such as predation or mutualism. The strength of graph theory lies in its ability to reveal systemic patterns, such as cycles, clusters, and networks of influence.

In systems theory, feedback loops—a critical concept—can be elegantly modelled using directed graphs. A **directed graph (digraph)** introduces directionality to edges, allowing us to capture cause-effect relationships. For example, in a heating system:

$$\mathbf{G} = (\{\mathbf{T}, \mathbf{H}, \mathbf{C}\}, \{(\mathbf{H} \rightarrow \mathbf{T}), (\mathbf{T} \rightarrow \mathbf{C}), (\mathbf{C} \rightarrow \mathbf{H})\})$$

where **T** is the temperature, **H** the heating element, and **C** the controller. The directed edges show how each component influences the next. Feedback dynamics, stability, and emergent behaviours become clear when visualized as graphs. Graphs also let us quantify systemic properties like **centrality** (identifying key nodes) or **path length** (measuring the efficiency of interactions). These insights are invaluable when analysing real-world systems like supply chains, social networks, or ecosystems.

While graph theory gives us a map of relationships, **category theory** elevates our understanding by focusing on the **transformations** between systems. If sets are the building blocks of mathematics, categories are the blueprints that describe how these blocks interact.

A **category** consists of:

- **Objects**, which represent systems or their components.
- **Morphisms** (arrows), which represent the transformations or processes between these objects.

For example, consider two enabling systems E_1 and E_2 , and a process p that transforms E_1 into E_2 . In category theory, this is written as:

$$E_1 \xrightarrow{p} E_2$$

The beauty of category theory lies in its abstraction. It allows us to model not only systems but also the **relationships between systems of systems**. This abstraction becomes invaluable when studying complex systems with layers of interactions.